

PHASE_8C_EXIT_GATE_EVIDENCE

Phase 8C — Exit-Gate Evidence Matrix (CLAUDE-1 Usability Proofs)

HXC-1613 | Sink-side gate document | Host OS: macOS darwin/arm64 | Audited 2026-06-03

Grounded in real code/tests under pkg/, internal/, cmd/, and the vasic-digital/security submodule (security/pkg/), cross-checked against the Phase 8C roadmap (docs/research/phase_08C/PHASE_8C_ROADMAP.md §8) and the code-grounded gap audit (docs/research/phase_08C/GAP_AUDIT.md).

Why this matrix exists (the sink-side gate)

Per **CLAUDE-1** (§7.1 + §11.4.39), tests are *necessary but not sufficient*. A green test on a non-functional feature is a **PASS-bluff** — the same defect class as a §7.1 violation. Phase 8C (the Chutes-AI remote-GPU integration) may **not** be declared *usable by an end user* on the strength of unit tests alone. This document is the **sink-side gate**: each Phase 8C exit gate (a CLAUDE-1 usability criterion drawn from roadmap §8) is mapped to the *end-user-visible behavior* it proves, the *evidence type* that proof requires, the *current status*, and a *pointer to a real test or the Queued ticket* that will close it.

Honesty rule (this IS the deliverable): a row is marked **PROVEN** only when a real, existing test/evidence file exercises the behavior on the host OS. Where only unit-level or record-layer proof exists (and the roadmap demands a live worker / real hardware / a higher-level flow), the row is **PARTIAL**. Where the implementation or its test is absent — *even if the registry marks the ticket “Completed”* — the row is **NOT-YET** and cites the Queued (or missing) ticket. The registry Completed flag is **not** accepted as evidence; only resolvable test files are.

Status legend: **PROVEN** = real test/evidence exists and runs on host · **PARTIAL** = real proof exists for a subset; roadmap-level gate (live worker / hardware / higher flow) not yet met · **NOT-YET** = no resolvable test/impl; tracked by a Queued ticket.

Evidence Matrix

Gate	End-user-visible behavior it proves	Evidence type required	Status	Pointer (real test / Queued ticket)
G1 — E2EE record interop	A confidential prompt sealed by one conforming impl is byte-identically reproducible and decryptable by another; tampering is rejected	Vector-driven interop test (deterministic AEAD)	PARTIAL	security/pkg/e2ee/interop/ciphertextltag vectors (record-labeled) §8 gate is “byte-for-byte vs <i>Chacha</i> Poly1305 cipher — both still open corpus, Queued), HXC-1561 (C) (AES ChaCha spec deviation)
G2 — E2EE envelope + framed transport	Tenant data travels as a length-prefixed encrypted record stream; replay/nonce-reuse/AAD/wrong-key are rejected	Real handshake + AEAD round-trip + rejection unit tests	PROVEN	security/pkg/e2ee/packageAAD/wrong-key); transport in s with round-trip + concurrent tes
G3 — E2EE streaming + discovery/nonce contract	Per-chunk SSE confidential streaming + single-use nonce / instance-discovery negotiation as an end user streams tokens	Live-worker streaming interop test	NOT-YET	HXC-1565 (streaming SSE E2E single-use nonce), HXC-1563 (Queued
G4 — Software proof-of-GPU (challenge/response)	A provider that does not hold the device secret cannot pass; expired/replayed/tampered/forged-descriptor responses are rejected	Real HMAC challenge/response rejection tests	PROVEN	security/pkg/gpuattest/pareplay/tamper/wrong-key/forged
G5 — PoVW spot-check (O(1)) + device-sealed encrypt	A large software-PoVW response is verifiable in O(1) at a	Spot-check + seal/open unit tests	PROVEN	security/pkg/gpuattest/se (TestSpotCheckValidIndexP TestSpotCheckTamperedSamp TestSpotCheckIsConstantWo TestSealOpenRoundTrip, Tes

Gate	End-user-visible behavior it proves	Evidence type required	Status	Pointer (real test / Queued ticket)
G6 — Hardware-rooted proof-of-GPU (real GraVal kernel)	random index; a single tampered word fails at that index; an AEAD key is bound to the device secret A spoofed / oversubscribed <i>physical</i> GPU is rejected; a genuine GPU passes; VRAM $\geq 95\%$ usable gate	End-to-end test on a real GPU node (GPU CI runner)	NOT-YET	Needs GPU + CUDA/OpenCL roadmap §7/§8. Tracked: HXC-1488 (HelixQA GraVal patch)
G7 — Software TEE attestation (quote construct/verify + policy)	An attestation document with a stale/forged nonce, clock-skew, or disallowed measurement is rejected; a conforming one verifies	ed25519-signed document construct/verify + policy unit tests	PROVEN	security/pkg/attestation/freshness / clock-skew / measurement seam is honestly self-labeled
G8 — Hardware TEE attestation (Intel TDX + NVIDIA quotes, measured boot, attested key release)	A tampered node fails admission; a genuine TDX+NVIDIA node releases the key/secret	Hardware integration test (real TDX + confidential GPU)	NOT-YET	Needs real TDX + H100/H200-DEFERRED per roadmap §7/§8 (construct/verify), HXC-1575 (Intel (NVIDIA NRAS/nvTrust), HXC-1576 (sek8s TDX+CC stack) — all QAs)
G9 — Cost-aware GPU scheduling	The scheduler prefers cheaper / freer GPUs; placement scoring is deterministic and tested	Scoring-plugin unit test	PROVEN	pkg/scheduler/cost_gpu_test price labels + proxy)
G10 — Attestation-gated scheduler admission	A spoofed / unattested node is filtered out <i>before</i> scoring; only an attested node is placeable; absent attestor fails closed	Two-node admission unit test (attested vs forged)	PROVEN	pkg/scheduler/attestation (TestSchedulerPlacesAttestation, TestSchedulerRejectsAttestation, TestFilterFailsClosedWithAttestation, TestSchedulerExpiredToken)

Gate	End-user-visible behavior it proves	Evidence type required	Status	Pointer (real test / Queued ticket)
G11 — Preemption + optimistic claiming	A higher-priority job preempts; concurrent claims resolve via version conflict (no double-placement)	Preemption + version-conflict unit test	PARTIAL	pkg/scheduler/gang_preempt_preemption + optimistic version multiplier model and SKIP_L0C (priority-based only)
G12 — GPU resource model (fingerprint + compute-multiplier catalog + attested flag)	A GPU is ranked by a real compute-multiplier catalog (H100 > A100 > default); an un-attested GPU scores below an attested one	Catalog ordering + scoring unit test	PROVEN	pkg/resources/gpuclass_test (ordering). Note: extended therm (Queued)
G13 — Model-integrity gate (hf_cache_verify → Go)	A tampered / truncated / missing model-cache file blocks serving; a clean cache is servable	E2E-style temp-dir manifest verify test (flip/truncate/delete)	PROVEN	pkg/modelintegrity/modelintegrity (TestVerifyFile_FlippedByTestVerifyFile_TruncatedIntegrityTestVerifyManifest_MissingTestVerifyManifest_OneTampered + size, V2 semantics)
G14 — LLMOrchestrator real inference (no stub)	Inference returns the REAL backend result and is rejected unless the model is registered AND loaded; empty prompt / missing backend are typed failures (never a fabricated string)	Backend-contract unit tests	PARTIAL	internal/llm/manager.go InferInference [...] string — it doesn't have guards. But the §8 live-serving is not met: HXC-1594 (vLLM/SGLang deploy), HXC-1472 (SGLang d
G15 — Confidential serving (relay sees only ciphertext)	A packet/log capture on the relay/proxy shows only ciphertext + routing metadata — the operator cannot	Packet/log capture (sink-side evidence) against a live confidential instance	NOT-YET	cmd/e2ee-proxy exists (HXC-evidence usability gate is open: encrypted-traffic gate), HXC-1472 — Queued

Gate	End-user-visible behavior it proves	Evidence type required	Status	Pointer (real test / Queued tick)
G16 — Scheduling throughput (≥ 10 placements/sec with attestation gate)	read tenant prompts Placement decisions sustain ≥ 10 /sec with the attestation filter engaged	Benchmark	NOT-YET	No throughput benchmark for the scheduler work; closed under scheduler perf work; closed under full request flow, Queued). handshake latency, not placement
G17 — Marketplace economic flow (registration $\rightarrow$ bounty $\rightarrow$ metering $\rightarrow$ payout)	A provider registers (capability proven), claims a bounty, serves metered invocations, and receives a computed payout	UUID-tagged e2e run with before/after evidence	PARTIAL	pkg/metering/metering.go ClaimBounty $\rightarrow$ Meter $\rightarrow$ Payout 9ad74be; registry evidence_pkg/metering_test.go TestClosure proves the end-to-end happy path bounty claim, exact unit accrual (payout); lifecycle enforcement (TestClosure2_LifecycleEnforcement (TestClosure3_ExactMetering) PROVEN. But the roadmap §8 <i>captured before/after evidence</i> artifact is not yet captured (closed). Mutation-coverage note: the record units*rate term) FAILED TestClosure1_RegisterClaim
G18 — Marketplace audit (commit-then-prove, reproducible reconciliation)	An audit report carries a SHA-256 commitment; per-node incentives are independently recomputed within tolerance	UUID-tagged reconciliation run with committed report	PARTIAL	pkg/auditproof/auditproof CommitReport takes a SHA-256 length-prefixed) serialization of compute-unit data; Verify index commitment and matches byte-registry evidence_path=pkg/auditproof_test.go proves i TestClause1_CommitThenProve TestClause2_TamperDetecte commitment), TestClause3_De Closure-level reproducible reconciliation gate is a <i>UUID-tagged reconciliation captured as evidence</i> — that since Mutation-coverage note: the record always-true) FAILED TestCla
G19 — Inference observability proxy (correlation-ID + anonymization)	Each request carries a propagated correlation-ID, the serving backend is recorded for audit, and configured fields are	Middleware unit tests	PROVEN	pkg/inferenceproxy/proxy_authorizer_test.go (correlation-keyed-hash anonymization detection)

Gate	End-user-visible behavior it proves	Evidence type required	Status	Pointer (real test / Queued ticket)
G20 — fiber miner^[2] validator transport	deterministically anonymized (stable per tenant) Two peers exchange length-prefixed frames with Ping/Pong keepalive over a single-reader/single-writer contract	Framed-transport round-trip unit test	PARTIAL	pkg/fiber/fiber_test.go p vs roadmap §9: signed-identity underlay not wired (documented)
G21 — New-pkg test coverage (>60% line)	Each new Phase 8C package carries genuine line coverage, not just smoke tests <i>Every</i> feature proven end-to-end for an end user with captured sink-side evidence; real integration (no mock-only) for E2EE/ attestation/ scheduling; HelixQA Challenge PASS reflects genuine usability, not a PASS-bluff	Codecov / go test -cover	NOT-YET	Coverage gate not yet asserted a chutes, pkg/marketplace, pkg/e2
G22 — CLAUDE-1 master usability gate		Captured screenshot/ log/metrics per feature + real-service/ hardware integration + HelixQA challenges	NOT-YET	The crypto/trust primitives (G2, PROVEN at unit/record level, b hardware (G6/G8), throughput evidence economic-flow runs (G PROVEN but whose live before open. HelixQA challenges Que HXC-1487 (multi-marketplace real-provider CI HXC-1554; us master gate stays NOT-YET un

Summary

- **PROVEN: 9** — G2, G4, G5, G7, G9, G10, G12, G13, G19 (each cites a resolvable test file that runs on the host)
- **PARTIAL: 6** — G1 (record-layer interoper only; Chutes-vector/ChaCha gate open), G11 (priority not value-multiplier; no SKIP LOCKED), G14 (real backend wired but no live vLLM/SGLang WRAP), G17 (closure-level economic flow PROVEN in pkg/metering; no UUID-tagged live captured-evidence run), G18 (closure-level commit-then-prove reconciliation PROVEN in pkg/auditproof; no UUID-tagged live captured-evidence run), G20 (transport only; no signed identity/E2EE underlay)

- **NOT-YET: 7** — G3, G6, G8, G15, G16, G21, G22 (each cites a Queued HXC ticket; G6/G8 are hardware/GPU-CI DEFERRED)

Grounding correction (supersedes a prior PASS-bluff guard): an earlier revision asserted G17 (HXC-1607) and G18 (HXC-1608) had *no implementing code/test in the tree*. That was a grounding error: it searched only pkg/marketplace for Bounty/Payout/Meter/Register and never consulted the registry item_history.evidence_path, which names the real implementation locations — pkg/metering for HXC-1607 and pkg/auditproof for HXC-1608, both landed in the same commit 9ad74be cited for the Completed marks. The flows DO exist and their closure-level behavior is proven (TestClosure1_RegisterClaimMeterPayout, TestClause2_TamperDetected, both passing on host). They are therefore **PARTIAL** (closure-level PROVEN; UUID-tagged live captured-evidence run still open), not NOT-YET. The registry Completed flag is still not by itself accepted as sink-side evidence — the upgrade to PARTIAL is justified by the resolvable, passing test files, not by the flag. The genuine residual concern is mutation coverage: the registry records a single FAILED mutation bite per gate (drop-units-term → TestClosure1; Verify-always-true → TestClause2), which is real but does not equate to code-nonexistence.

Bottom line: Phase 8C's trust/confidentiality *primitives* are genuinely proven at the unit/record layer on the host, and the marketplace economic + audit flows now have proven closure-level implementations (G17/G18 PARTIAL). Phase 8C is still **NOT yet declarable end-user-usable** (G22 NOT-YET): the live-worker, real-hardware, throughput, and the UUID-tagged captured-evidence economic/audit runs remain open and tracked by Queued tickets.

Evidence file index (every PROVEN-row path)

- security/pkg/e2ee/package_test.go, security/pkg/e2ee/transport.go — G2
- security/pkg/gpuattest/package_test.go — G4
- security/pkg/gpuattest/seal_test.go — G5
- security/pkg/attestation/attestation_test.go — G7
- pkg/scheduler/cost_gpu_test.go — G9
- pkg/scheduler/attestation_test.go — G10
- pkg/resources/gpuclass_test.go — G12
- pkg/modelintegrity/modelintegrity_test.go — G13
- pkg/inferenceproxy/proxy_test.go, pkg/inferenceproxy/authorizer_test.go — G19
- security/pkg/e2ee/interop_test.go — G1 (PARTIAL: record-layer only)
- pkg/scheduler/gang_preempt_test.go — G11 (PARTIAL)
- internal/llm/manager.go — G14 (PARTIAL)
- pkg/metering/metering.go, pkg/metering/metering_test.go — G17 (PARTIAL: closure-level flow proven; no UUID-tagged live run)
- pkg/auditproof/auditproof.go, pkg/auditproof/auditproof_test.go — G18 (PARTIAL: closure-level reconciliation proven; no UUID-tagged live run)
- pkg/fiber/fiber_test.go — G20 (PARTIAL)